

Datathon 2026 - Team Report

Team Name: 5Gen data

Team Members:

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Track(s) addressed: Track 1 and 3

1. Problem Statement

There is a significant gap in specific mathematical competencies like Measurement, Division and Place value in Karnataka. This varies by district and is partially explained by Socio-economic factors at district level schools. This leads to the suggestion of competency-targeted rather than generic interventions.

2. Data and Methodology

2.1 Primary dataset

Primary dataset used for analysis was directly collected using the Intranet provided by the organisers. The source contains data from a math contest of grades 4,5 and 6 of academic year 2022-23 from Karnataka (district level data). The Assessment Data contains columns such as State, District, Block, Cluster, GP Name, GP ID, Gender, Questions 1 to 20 and Unique Identifier. Competency Mapping spreadsheet consists of columns such as Questions, Question Name and Competency. District and GD Mapping spreadsheet consists of columns such as State, District, Block, Cluster, GP Name and GP ID. There are entries of 312,702 students from grades 4-6, 20 competency questions, 30 districts.

2.2 External datasets used

The external dataset used for analysis is from NFHS-5 district factsheets. Source: <https://www.data.gov.in/catalog/national-family-health-survey-5-nfhs-5-india-districts-factsheet-data-provisional>. The external dataset contains columns such as child nutrition, anaemia and female literacy.

2.3 Cleaning steps

1. The primary dataset provided was pre-cleaned by Akshara Foundation and there were no incorrect or missing values which were present in the dataset.
2. The dataset had three grade-level files which were combined into a single dataset
3. For cross-dataset analysis, district names were harmonised to the pre-2014 naming convention used in NFHS-5 and two education sub-blocks were combined into their parent districts. “Bellary (proxy for Vijayanagar, which was created in 2021, after the survey period for NFHS-5) was the pre-split parent district of Vijayanagar, a limitation we explicitly state.

2.4 Analytical approach

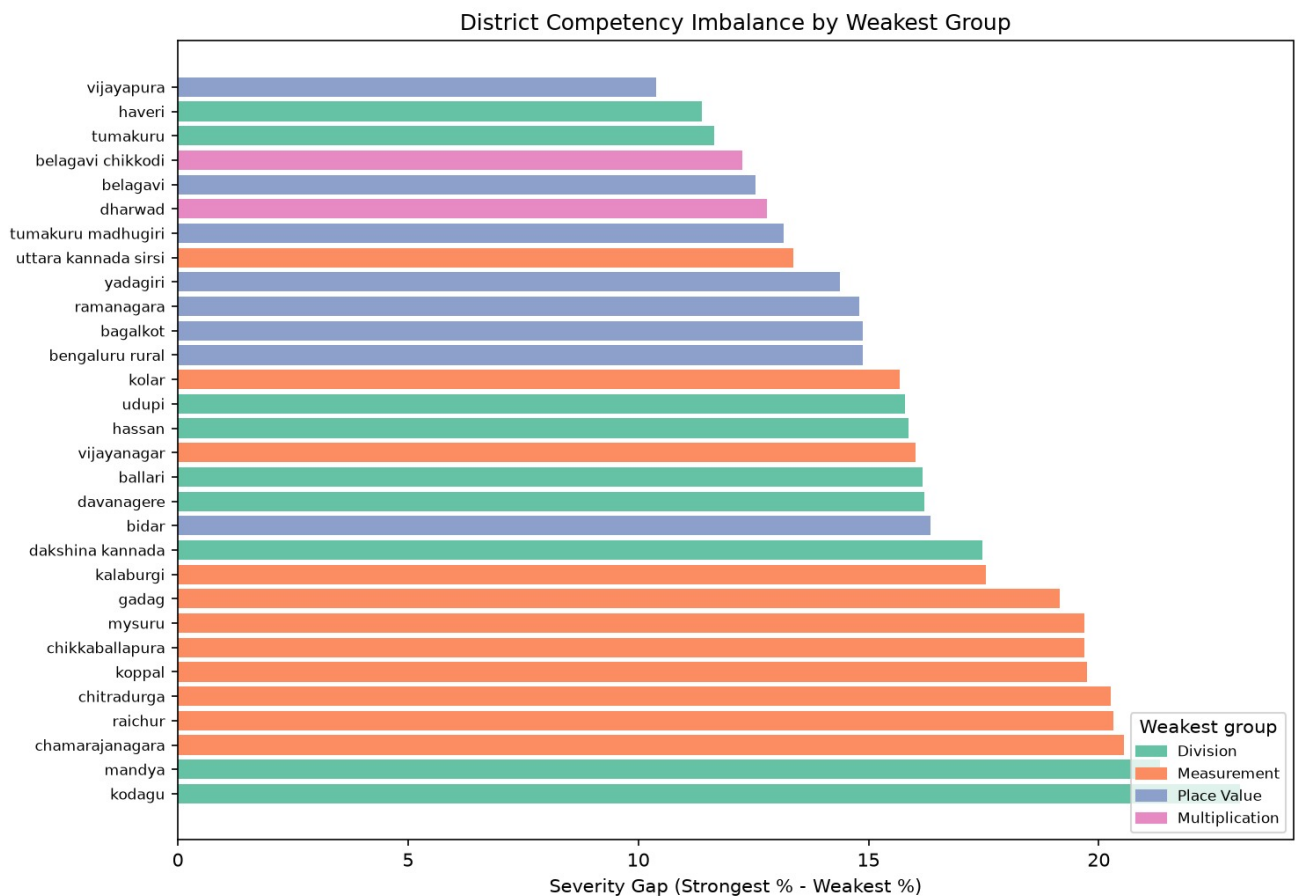
1. Questions 1 to 20 were grouped into 9 competency categories using the official competency mapping which was computed per-district, per-competency accuracy.
2. We identified each district's weakest competency and the severity gap (The weakest vs strongest competency within that district).
3. The competency weaknesses identified are grouped into three clusters namely Measurement (11 districts), Division (9 districts), Place Value (8 districts) The large districts (Kalaburgi, Chitradurga, Raichur, Koppal) are concentrated in to Measurement-weak cluster.
4. District-level results linked with NFHS-5 (female literacy, school attendance) using a corrected district-name crosswalk (accounting for 2014 district renamings).
5. Found: Female literacy is a moderate correlate of a district's overall competency floor ($r \sim 0.45$) but not of how uneven a district's profile is ($r \sim 0.0-0.16$) – distinction

between “general disadvantage” districts and “specific instructional gap” districts.

3. Key Findings

Finding 1

Measurement, Division and Place Value are the three most common weakest competencies across the state which accounts for 28 out of 30 districts.



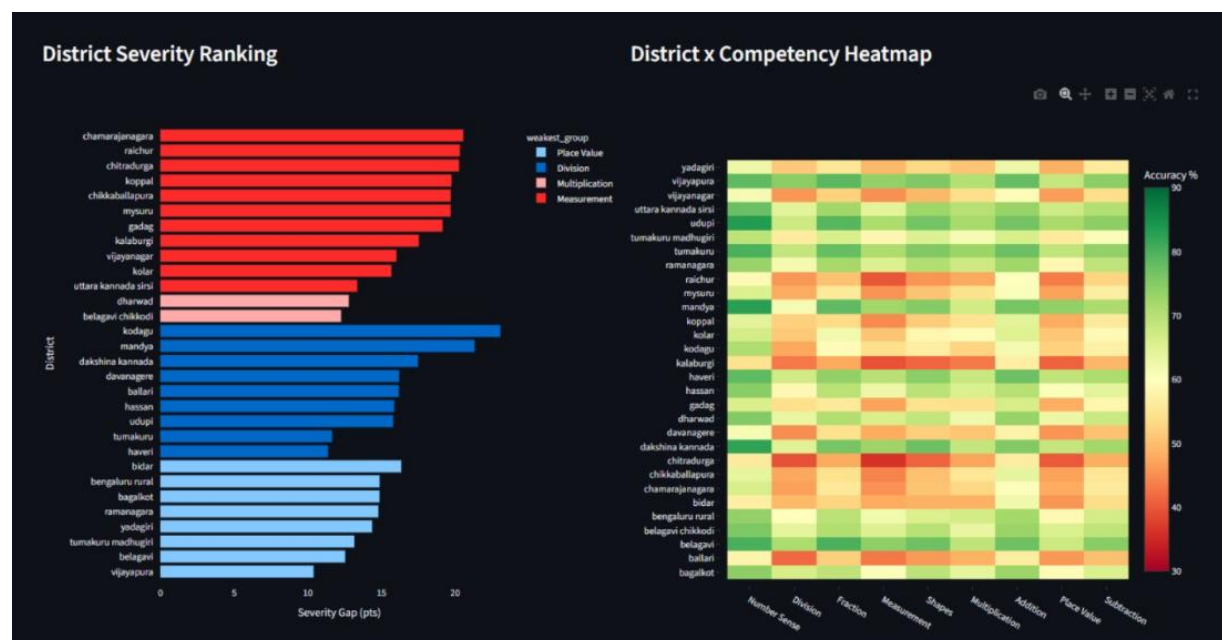
Why this matters: The importance lies in the fact that this is not random noise but a focused and identifiable pattern. Since almost every district's weakest aspect falls into

just three areas of competence, the state does not need thirty separate remediation plans; instead, it only needs three properly designed instructional modules which are then deployed selectively depending on which of the three areas the district falls into.

Decision to be taken: Education department has to develop (or obtain) standard teaching programmes for these three skills at this stage instead of delaying them until a district-by-district diagnosis has been carried out as the diagnostic work is mostly already completed.

Finding 2

Gap severity depends on which skill is lacking (Kodagu's 23-point division gap vs Vijayapura's 10-point gap, given Vijayapura's lower literacy).

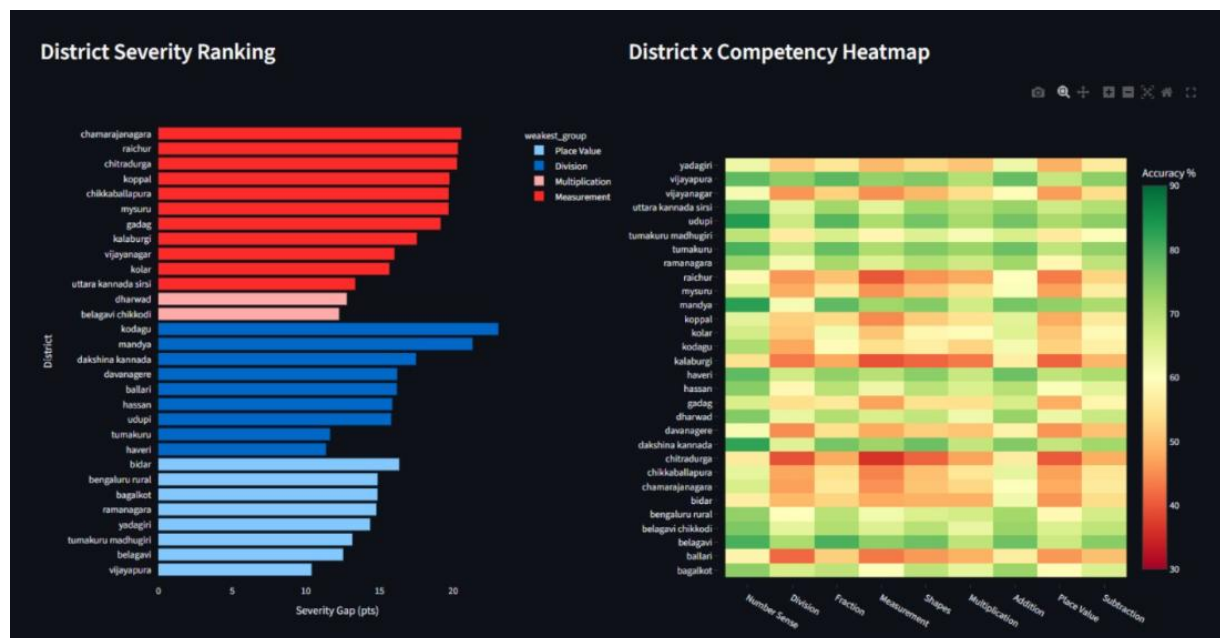


Why this matters: This separates “how bad” from “what’s wrong” and shows that severity is not driven by poverty or literacy but a pedagogical one.

Decision to be taken: High severity, high literacy districts require narrow retraining for teachers to fill the specific skill gaps and should definitely not be combined with other, large “disadvantaged district” initiatives as that offers solutions to problems that do not exist in those districts.

Finding 3

Socio-economic status and the performance floor of a district show a positive correlation ($r \sim 0.45$). However, socio-economic status and the internal imbalance of a district are weakly correlated ($r \sim 0.0$ to 0.16).



Why this matters: This is the main takeaway from the analysis as they show that when problems are combined, they end up allocating interventions in the wrong place. Districts that are on the low end of the performance floor require more support in the areas of literacy, attendance and available resources. Districts that have an unreachable performance floor but large amount of internal imbalance requires a completely different approach (competency training).

Decisions to be taken: Each district should be evaluated against two questions before assignment: does the district's overall performance have a substantial gap and if so, what's the weakest competency? Multiple interventions would be required to assist a district that is low on the performance floor. For districts that have high imbalance at a sufficient floor competency, target interventions are required that relate specifically to the field of teaching.

4. Recommendations

1. Teachers in Kalaburgi, Chitradurga, Raichur and Koppal should receive special training on teaching Measurement. This is because these districts have the most students and consistently score low in Measurement. Helping more children with this skill will make a big difference.
2. The GKA Program Team should give extra practices and teaching help for Division in districts like Kodagu and Mandya. Students there mostly struggle with Division, even though they generally do well in school. Focusing help on this specific area should work better than trying to help with everything.
3. The District Administration and Akshara Foundation should act on the need to improve basic learning in Raichur and Yadgir. This means offering things like help with basic maths, mentoring for teachers and better learning materials. These districts have broader learning problems, often due to lower socio-economic factors, so they need all-around support, not just help with one math topic.